

**Exam Shield**

Offline Handwritten Evaluation Auditor

Ensuring Academic Integrity & Computational Sum Checks Before Results
Publication

HACKATHON TRACK: COMPUTER VISION

SNIST - HACK-THE-MATRIX 2026

LOCAL & OFFLINE ARCHITECTURE

1. The 30-Second Elevator Pitch

The Pitch Concept

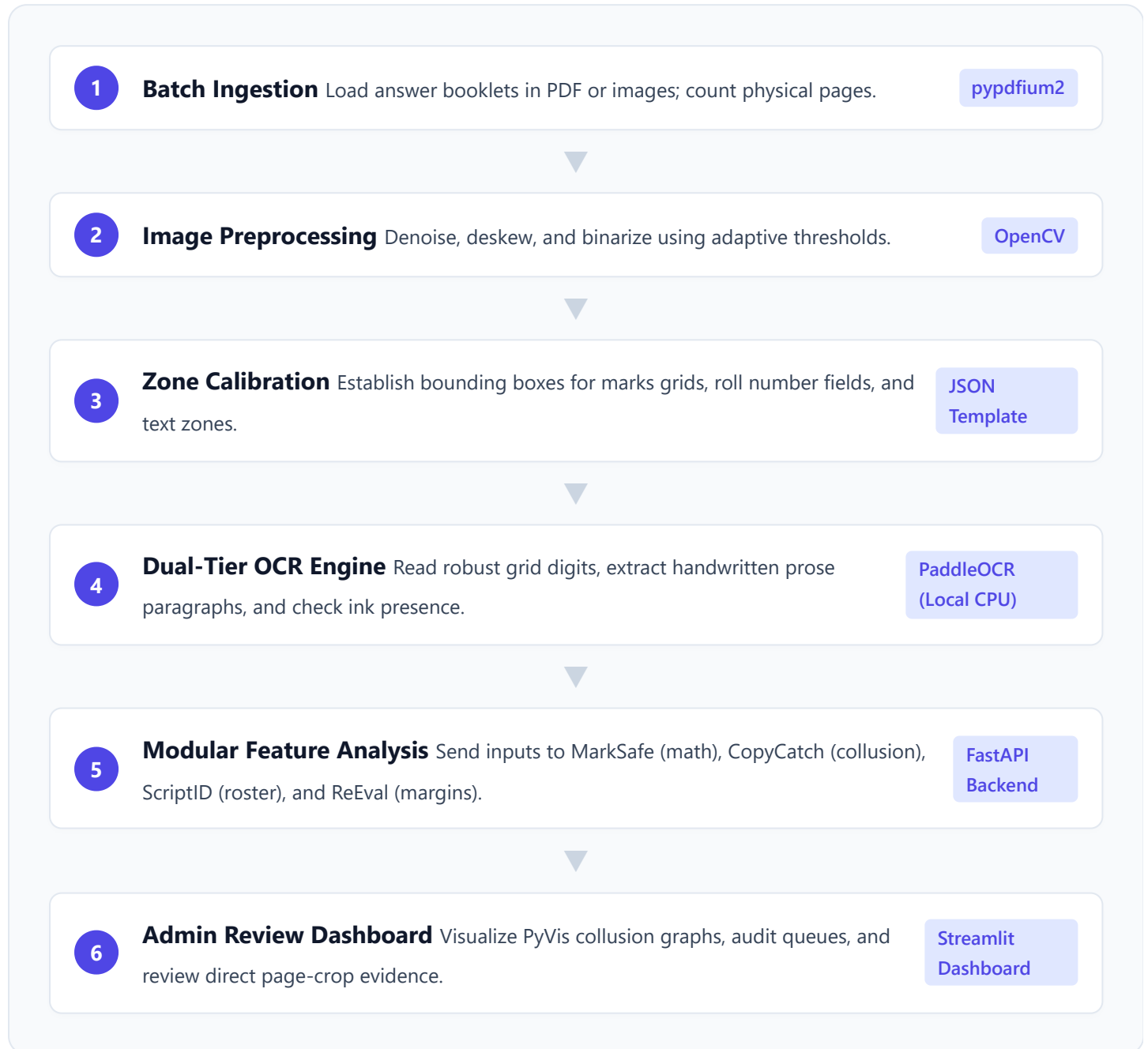
Every semester, academic institutions manually grade hundreds of thousands of handwritten scripts. This manual chain of trust suffers from transcription errors, summing slips, and student collusion networks that are invisible to a grader checking pages sequentially. **ExamShield** is a local computer vision assistant for exam evaluation cells. It reads digit scoring blocks, validates cover page additions, maps student collusion graphs via prose semantic analysis, and flags borderline scores for double checking—all **offline** on standard CPUs, maintaining total data privacy.

2. The Problem Space (Legacy vs. ExamShield)

Workflow / Category	Legacy Manual Workflow	 ExamShield Automated Audit
Collusion & Plagiarism	0% Coverage. Graders evaluate scripts one-by-one. In a 300-paper cohort, cross-referencing involves 44,850 pairs —which no human evaluator can complete.	100% Coverage. Digitizes handwritten prose, computes similarity metrics, and maps student copying clusters in seconds.
Arithmetic Audits	Manual Totals Review. Staff manually check math calculations across cover grids, leading to transcription slips and omitted question audits.	Automated Calculations. MarkSafe parses OCR digits, verifies sums, and alerts administrators of math mismatches automatically.
Grade Boundary Disputes	Post-Publication Backlogs. Students scoring just below thresholds (e.g. 39/100, passing score = 40) file retroactive paid revaluations.	Preemptive Audits. ReEval Guard flags boundary margins, queuing them for checks <i>before</i> final scores are locked and published.
Roster Accuracy	Manual Keying Errors. Administrative typing errors attribute grades to wrong students, resulting in enrollment scandals.	Registration Verifier. ScriptID matches digit OCR roll numbers against class lists to flag absentees and duplicates instantly.

3. Shared Ingestion Spine & Architecture

Every feature functions along a unified local pipeline, routing document scans sequentially into specific validation blocks.



4. Core Evaluation Engines (Deep Dive)

MarkSafe Engine

Trust Layer

PaddleOCR Digits + OpenCV Pixel Density

- Extracts question marks and cover page totals from predefined templates.
- Parses complex tokens using regular expressions, handling integers, decimals, fractions (e.g., $\frac{9}{10}$), and evaluable additions (e.g., $4+3.5$).
- **Safe Ambiguity Handling:** Skipped and low-confidence markings (OCR confidence < 0.85) are marked `AMBIGUOUS_MARK` and routed to a reviewer instead of making false guesses.
- Triggers `SUM_MISMATCH` if question sums fail to match the written overall total.

CopyCatch Engine

Headline Demo

Prose OCR + sentence-transformers (all-MiniLM-L6-v2)

- Converts handwritten answer prose to vector embeddings locally.
- **Class-Baseline Anomaly Ranking:** Discards class-wide noise (common blackboard formulas, definitions) by computing z-scores: $Z_{\{A,B\}} = \frac{\text{Similarity}(A,B) - \mu_{\{\text{class}\}}}{\sigma_{\{\text{class}\}}}$.
- Links collusion paths only if $Z_{\{A,B\}} \geq 3.0$ (representing standard deviations away from class average).
- Generates interactive PyVis HTML node graphs. Graders click graph edges to inspect side-by-side student handwriting.

ScriptID Engine

Roster Auditor

Digit OCR + pandas Registration Lookup

- Processes the student Roll Number zone on answer booklet covers.
- Cross-references entries against class register database tables in CSV formats.
- Flags `UNREGISTERED_ID` (OCR adjustments are easy via visual cover crops) and `DUPLICATE_ID` (if multiple booklets share a roll number).
- Flags `ABSENTEE_WITH_SCRIPT` to protect institutions from exam-hall envelope logging leaks.



ReEval Guard Engine

[Dispute Reducer](#)

Pure Python Logic Overlay

- Scans final scores against institutional grade cutoffs and pass-fail lines.
- Identifies scripts located within border margins (e.g. 1-2 points below passing thresholds).
- Adds scripts to high-priority review pipelines to resolve math checks prior to final score locking.



BlankCheck & RubricLens (Stretch Features)

[Assist Engines](#)

OpenCV Density + cross-encoder/nli-deberta-v3-xsmall

- **BlankCheck:** Runs pixel density checks to mark blank pages, preserving physical workload history.
- **RubricLens:** Uses Natural Language Inference to check paragraphs against rubrics. Highlight outputs show **Entailment (Green)** or **Contradiction (Red)**. *It never awards grades—the human decides.*

5. Technical Stack & Local-Execution Justifications

Designed for air-gapped exam rooms, our stack avoids external dependencies and runs efficiently on standard laptop hardware.

Stack Layer	Technology Choice	Why it Wins the Judge Q&A
Image & CV Spine	OpenCV, pypdfium2, Pillow	Performs page extraction, deskewing, and image binarization instantly, with no licensing cost.
Local OCR Engine	PaddleOCR (CNN-based)	Offline, free, and yields character confidence levels—vital for checking scribbled or crossed-out scores.
Semantic Vectors	sentence-transformers (all-MiniLM-L6-v2)	Lightweight (~80 MB) and runs vector calculations on standard CPUs in fractions of a second.
NLI Inference	cross-encoder/nli-deberta-v3-xsmall	Handles negation (e.g., flagging "is not exothermic" as a contradiction) which keyword searches miss completely.
Dashboard UI	Streamlit & PyVis (vis.js)	Constructs interactive graphs and audit lists in hours, bypassing React development overhead.

6. Live Demo Choreography

00:00 -
00:30

The Core Hook

"Judges, graders checking papers sequentially cannot see batch-wide networks. ExamShield finds what humans miss."

00:30 -
01:30

The Live Interactive Scan (Judges are the Data)

Judges write responses on physical sheets. Two judges copy each other. The team photographs the papers and uploads them live.

01:30 -
02:30

CopyCatch Collusion Map Update

The dashboard collusion graph updates live. The nodes representing the two copying judges connect with a red line. Clicking the link reveals side-by-side text overlaps.

02:30 -
03:00

MarkSafe cover-page error display

Load a pre-configured 35-paper test corpus. Show cover calculation mismatches and hand-drawn alterations marked "AMBIGUOUS". Show how easily an auditor overrides them.

7. Q&A Defense Playbook (Judge Prep)

Q: OCR on handwriting is noisy. How do you avoid printing incorrect grades?

A: We operate under a strict human-in-the-loop policy. The system never inputs final marks automatically. If PaddleOCR confidence score drops below 85% due to a strikeout or scribble, MarkSafe locks it as **AMBIGUOUS** and prompts visual cover-page card reviews.

Q: Pairwise text comparison is $O(N^2)$. How does this scale for large batches?

A: Text is digitized and vectorized once. Cosine similarity comparisons over 1,000 papers (10^6 operations) on a standard CPU require less than 1.5 seconds using optimized NumPy matrix calculations, making scaling highly manageable.

Q: What if two students write similar answers because they memorized the same lecture slide?

A: This is why we implemented Class-Baseline Anomaly Ranking. If the whole cohort shares similar wording, the baseline mean (μ_{class}) rises. CopyCatch maps z-scores; links only light up if student pairs diverge significantly from the class-wide average.

Q: Why not use commercial cloud models like GPT-4o?

A: Examination data is highly confidential and cannot be sent to third-party servers. Additionally, local CNNs (PaddleOCR) and sentence vectors (MiniLM) are completely free, run offline in secure cells, and supply character coordinates and confidence boundaries that cloud models hide.